ESWARI L R



Junior AI Engineer | Machine Learning & Python Developer

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Portfolio | LinkedIn | GitHub | LeetCode

Summary

Pre-final year Information Technology student with solid foundations in Python, Machine Learning, Data Structures, and API development. Experienced in building computer vision models with YOLO and OpenCV, data modeling using Pandas and NumPy, and deploying real-time voice-driven AI interfaces using FastHTML and REST APIs.

Technical Skills

Programming Languages: Python, Java, JavaScript, SQL
Web Development: React, HTML, CSS, Flask, REST APIs
AI, ML & Computer Vision: Machine Learning, OpenCV, YOLO, Prompt Engineering
Data Science & Analytics: Pandas, NumPy, Scikit-Learn, Matplotlib, Data Preprocessing
Tools & Platforms: Git, GitHub, Docker, AWS, Postman, VS Code, Render, MySQL

Projects

Guardian Eyes | AI PPE Safety Compliance Monitoring System GitHub
Python, OpenCV, YOLO, Computer Vision, Real-Time Inference
- Trained a YOLO model with OpenCV for real-time safety gear (helmets, vests) detection in industrial settings.
- Implemented bounding-box inference pipelines and violation alert triggers, reducing manual surveillance needs.

DropSaviors | Predictive Rainwater Harvesting Assessment Engine GitHub
Python, Flask, Scikit-Learn, Pandas, NumPy, Geospatial Analytics
- Developed an analytical engine modeling rainwater runoff potential using geospatial data and rainfall metrics.
- Built modular Flask REST API endpoints to serve real-time runoff scores and interactive visual analytics.

Voice AI Agent | Real-Time Conversational Interface GitHub
Python, FastHTML, Ultravox Realtime API, REST APIs, WebSockets
- Built a real-time conversational voice agent leveraging the Ultravox API for low-latency speech-to-speech interaction.
- Designed asynchronous endpoints handling call sessions, live transcript streaming, and deployed the app on Render.

Experience

Java Full Stack Developer Intern | *Next Future Technology Pvt. Ltd.* Mar 2026
On-Site | Trichy, India
- Implemented backend modules in Java utilizing OOP principles and Collections Framework for scalable data flow.
- Analyzed runtime efficiency, debugged algorithmic bottlenecks, and enhanced core application stability.

Web Development Intern | *Coding Samurai* Mar 2025 – Apr 2025
Remote
- Developed responsive client interfaces and maintained structured version control workflows via Git and GitHub.

Education

Kongunadu College of Engineering and Technology 2023 – 2027
Bachelor of Technology in Information Technology **CGPA: 8.45 / 10.0**

Sacred Heart Girls Higher Secondary School 2021 – 2023
Higher Secondary Certificate (HSC) **Percentage: 90.83%**

Achievements & Competitions

- **1st Prize:** Code Debugging & Paper Presentation at COMBLAZE 2K25 technical symposium.
- **1st Prize:** HACKNOVA 2K26 Hackathon; **3rd Place:** GENESIS'26 24-Hour Hackathon.
- **Participant:** Shaastra AI/ML Challenge Techathon (IIT Madras) & AMD Slingshot Hackathon.
- **Published Author:** Research paper *"Smart Health Monitoring and Medication Reminder Application"* in IJSRET.